



A Novel Convolutional Neural Network–Long Short-term Memory Model for Interplanetary Coronal Mass Ejection Detection

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Abstract

Current models for detecting interplanetary coronal mass ejections (ICMEs) using spacecraft in situ measurements rely heavily on manual processes, resulting in time-consuming procedures, inconsistencies in identification criteria, and gaps in existing catalogs. Although several machine learning–based detection models have been proposed, there is still a need for improved accuracy. This study presents a convolutional neural network–long short-term memory (CNN-LSTM) model with dynamic loss function, designed to efficiently process complex multidimensional spatiotemporal data. The model leverages LSTM to capture evolving patterns in time series data, while employing a CNN to extract spatial features, making the model highly adaptable. Using in situ data collected by the Wind spacecraft from 1997 October 1 to 2016 January 1, the proposed model demonstrates strong competitiveness. Trained on data from 1998 January 1 to 2009 December 31, 189 out of 230 ICMEs are detected in the test set, achieving an F1 score of 81.29%.

Unified Astronomy Thesaurus concepts: [Astronomy data analysis \(1858\)](#); [Computational methods \(1965\)](#)

1. Introduction

Coronal mass ejections (CMEs) are spectacular phenomena involving the eruption of magnetized plasma from the surface of the Sun into the heliosphere (J. P. Byrne et al. 2010). Interplanetary coronal mass ejections (ICMEs) are in fact a manifestation of CMEs in interplanetary space. Interacting with the solar wind leads to changes in ICME velocity (A. Borgazzi et al. 2009), as well as alterations in their structure, magnetic field, and plasma properties (L. Burlaga et al. 1981; L. Klein & L. Burlaga 1982; A. J. Weiss et al. 2021). Magnetic clouds (MCs), a subclass of ICMEs recognized for their strong geoeffectiveness, are thought to be the main drivers of geomagnetic disturbances and severe space weather on Earth and other planets in the solar system (T. Žic et al. 2015). Figure 1 shows the typical in situ measurements of an MC recorded by the Wind spacecraft (K. Ogilvie & M. Desch 1997) between 1997 November 6 and 1997 November 10, where the dashed lines indicating onset and termination times. The top panel shows an enhanced magnetic field compared to the surrounding ambient solar wind, along with a long and smooth rotation of its components. This is related to the lower proton temperature (third panel), resulting in lower values of the plasma parameter β , which is defined as the ratio of thermal pressure to magnetic pressure (fifth panel). The second panel shows that the speed has increased compared to previous solar winds and is on a declining trend. The characteristics of MCs are simultaneous sudden increases in magnetic field strength and velocity (as shown on the left side of the dashed line in the top panel), and there is a sheath, which is the turbulent region between the

shock and the MC. These are the main criteria typically used to identify ICMEs (Y. Chi et al. 2016).

The deployment of the Wind spacecraft (K. Ogilvie & M. Desch 1997) and Advanced Composition Explorer (ACE; E. C. Stone et al. 1998), which monitor the near-Earth solar wind, has enabled comprehensive studies that have advanced the cataloging of ICME events. L. Jian et al. (2006) meticulously documented 230 ICMEs occurring between 1995 and 2004, and delved into the periodic variations of parameters such as the shock association rate and scale size, thereby shedding light on the dynamic nature of these events. Building upon this, a comprehensive survey by L. K. Jian et al. (2008) analyzed 124 ICMEs spanning from 1979 to 1988 and found that the occurrence of magnetic-cloud-like ICMEs decreases as solar activity increases. This provides valuable insights into the relationship between solar dynamics and ICME occurrences. R. P. Lepping et al. (2006) visually identified 82 MCs from 1995 to 2003, summarizing various findings, including methods for identifying the front boundary of an MC. I. Richardson & H. Cane (2010) further enhanced the understanding of ICMEs by presenting an updated catalog comprising approximately 300 near-Earth ICMEs observed from 1996 to 2009. This catalog integration included additional considerations such as solar wind composition and charge-state observations for precise identification. Y. Chi et al. (2016) compiled a comprehensive catalog covering 465 ICMEs from 1995 to 2015, focusing on ICME-driven shocks and the modulation of ICME-related parameters in correlation with the solar cycle. T. Nieves-Chinchilla et al. (2018) listed 337 ICMEs spanning three solar cycles from 1995 to 2015, with 298 of these displaying organized magnetic field signatures. Notably, Y. Chi et al. (2016) observed that, on average, 80% of the ICMEs in a given list exist in another list.

In recent years, the integration of machine learning and deep-learning models has shown promise in solar physics



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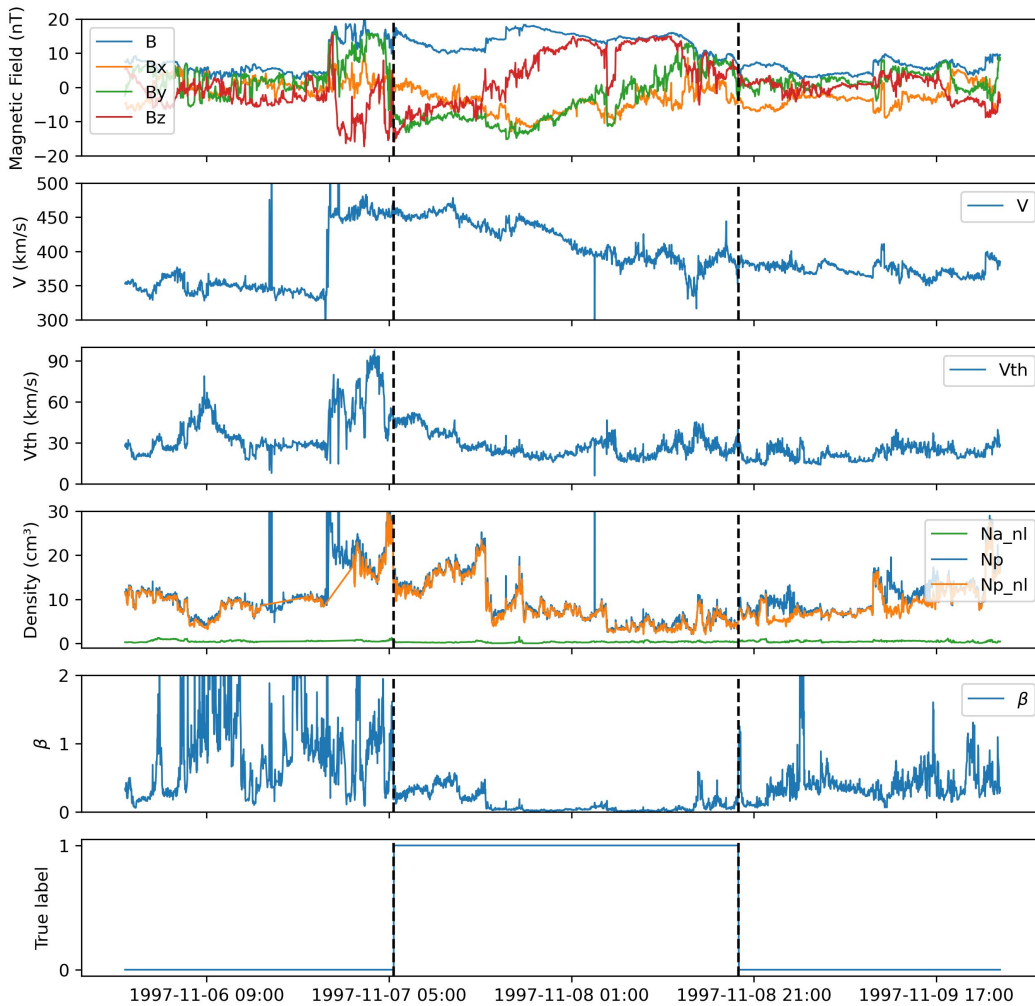


Figure 1. Observations of solar wind dynamics during an ICME by the Wind spacecraft at the L1 Lagrange point in 1997. The vertical dotted lines mark onset and termination times of an ICME event. Panels from top (1) to bottom (5) show: (1) magnetic-field-related features B , B_x , B_y , and B_z ; (2) solar wind velocity V ; (3) thermal velocity V_{th} ; (4) proton and alpha particle densities obtained from the analysis of moments and nonlinear analyses N_p , $N_{p,nl}$, and $N_{\alpha,nl}$, respectively; (5) plasma parameter β ; (6) event labeling.

(A. Asensio Ramos et al. 2023), such as flare classification (F. Benvenuto et al. 2018; K. Florios et al. 2018; M. K. Georgoulis et al. 2021; F. Ribeiro & A. Gradwohl 2021; S. Sinha et al. 2022), sunspot-number prediction (Z. Pala & R. Atici 2019; S. S. R. Moustafa & S. S. Khodairy 2023) and ICME identification (L. dos Santos et al. 2020; T. Narock et al. 2024; S. Pal et al. 2024). In ICME forecasting, M. M. Mahdi et al. (2019) employed an ensemble bagged tree model (L. Breiman 1996), achieving an accuracy of 90.8%. G. Nguyen et al. (2019) developed multiple convolutional neural networks (CNNs; Y. LeCun & Y. Bengio 1998) to predict ICME events from 1997 to 2015, achieving a recall rate of 84% and a precision of 84%. J. Chen et al. (2022) further advanced deep-learning-based forecasting by integrating residual modules (K. He et al. 2016) into U-net (O. Ronneberger et al. 2015) to tackle ICME prediction as a time-series segmentation problem, resulting in an F1 score of 80.18%. Similarly, H. T. Rüdiger et al. (2022) developed the ResUNet++ model (D. Jha et al. 2019) to improve ICME detection efficiency, achieving a true skill statistic of 0.64 on Wind spacecraft data from 1997 to 2015.

While prior research has made some progress, models struggle to effectively capture long-term temporal patterns, which poses a critical obstacle to the accurate prediction of ICMEs. To address this issue, long short-term memory (LSTM; S. Hochreiter & J. Schmidhuber 1997), a specialized recurrent neural network architecture, is introduced. Thanks to its unique memory cells and gating mechanisms, LSTM can effectively handle long-term dependencies, overcoming the limitations of traditional models in capturing such extended temporal patterns. Building on this, our study incorporates LSTM and combines it with the spatial feature extraction capabilities of a CNN to propose a CNN-LSTM-based ICME-forecasting model, aiming to enhance detection accuracy and robustness.

The rest of the paper is organized as follows. Section 2 introduces the data and offers a comprehensive overview of the pipeline. Following that, Section 3 offers a detailed analysis of the experiments conducted. Section 4 offers relevant discussions. Finally, Section 5 summarizes the findings, draws conclusions, and discusses potential avenues for future research.

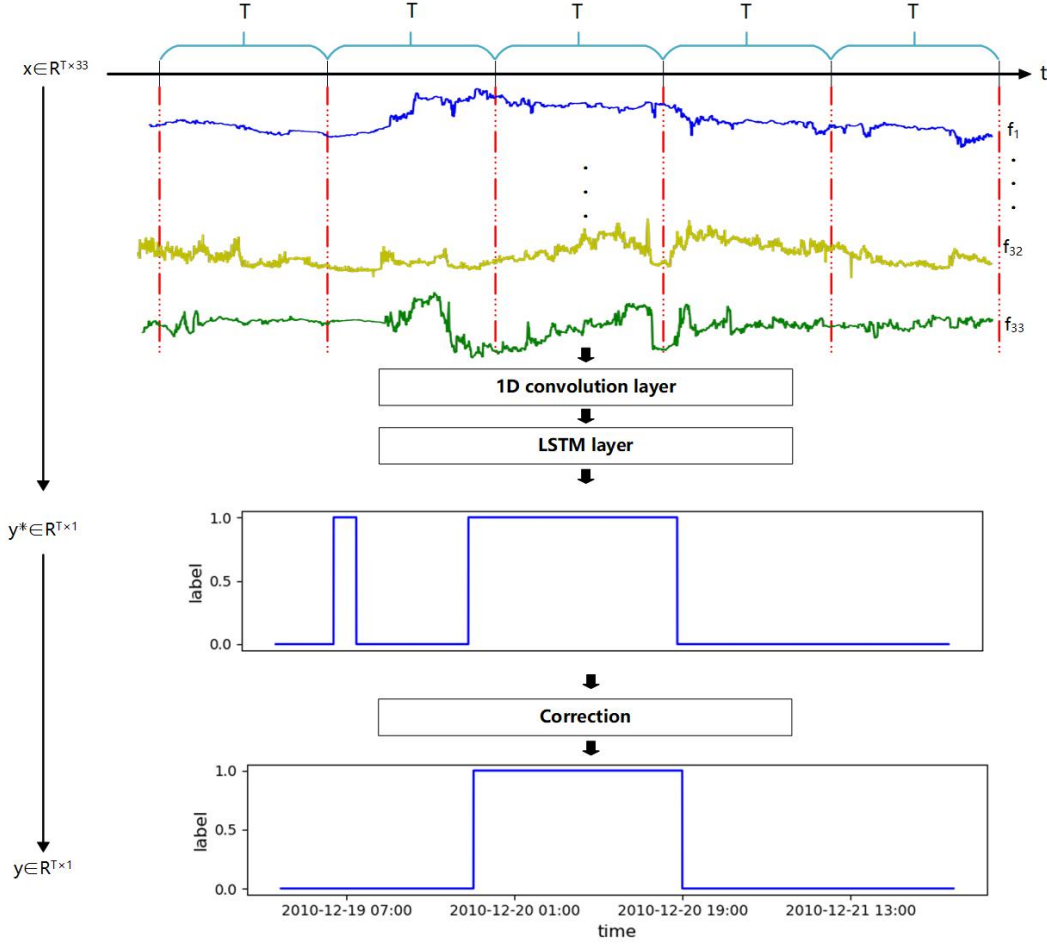


Figure 2. Workflow of the proposed CNN–LSTM model for ICME detection. Input a temporal segment $x \in R^{T \times 33}$; raw/corrected outputs $y^*, y \in \{0, 1\}$ with 1 indicating ICME events.

2. Data and Pipeline

In this section, we introduce the data and provide a detailed description of the ICME detection process. The workflow is illustrated in Figure 2. The preprocessed data is split into segments having an equal number of data points and then fed into a hybrid CNN–LSTM model. During training, a dynamic loss function is employed to mitigate class imbalance and a correction strategy is used to refine detection accuracy.

2.1. Data

Data description. The ICME data used in this paper was collected from 1997 October 1 to 2016 January 1, by three instruments on board the Wind spacecraft: the Magnetic Field Investigation (R. Lepping et al. 1995), Solar Wind Experiment (K. Ogilvie et al. 1995), and 3D Plasma and Energetic Particles Experiment (R. Lin et al. 1995). The ICME catalog was compiled from five existing catalogs (L. Jian et al. 2006; R. P. Lepping et al. 2006; I. Richardson & H. Cane 2010; Y. Chi et al. 2016; T. Nieves-Chinchilla et al. 2018), encompassing 653 ICME events. Both the data and ICME catalog are available at <https://github.com/gautiernguyen/Automatic-detection-of-ICMEs-at-1-AU-a-deep-learning-approach>.

The above five catalogs collectively contain a total of 5,482,361 data points, with each data point comprising 33 features. As shown in Table 1, the primary 30 features

Table 1
Description of Data Characteristics

Feature	Unit	Description
V	km s^{-1}	Bulk velocity
V_x, V_y, V_z	km s^{-1}	Bulk velocity components
V_{th}	km s^{-1}	Thermal velocity
B	nT	Magnetic field
B_x, B_y, B_z	nT	Magnetic field components
σ_B	nT	Normalized magnetic fluctuations $\sigma_B = \sqrt{\sigma_{B_x}^2 + \sigma_{B_y}^2 + \sigma_{B_z}^2} / B$
$\sigma_{B_x}, \sigma_{B_y}, \sigma_{B_z}$	nT	rms of magnetic field components
N_p	cm^{-3}	Density of protons obtained from moment analysis
$N_{p,nl}$	cm^{-3}	Density of protons obtained from nonlinear analysis
$N_{\alpha,nl}$	cm^{-3}	Density of α particles obtained from nonlinear analysis
P_{dyn}	nPa	Dynamic pressure $P_{dyn} = N_p V^2$
β	...	Ratio between the thermal and the magnetic pressure
$F_0 \sim F_{14}$	$\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1} \text{keV}^{-1}$	15 channels of proton flux

encompass the bulk velocity V and its components V_x, V_y, V_z ; the thermal velocity V_{th} ; the magnetic field B and its components B_x, B_y, B_z , along with the corresponding rms of the magnetic

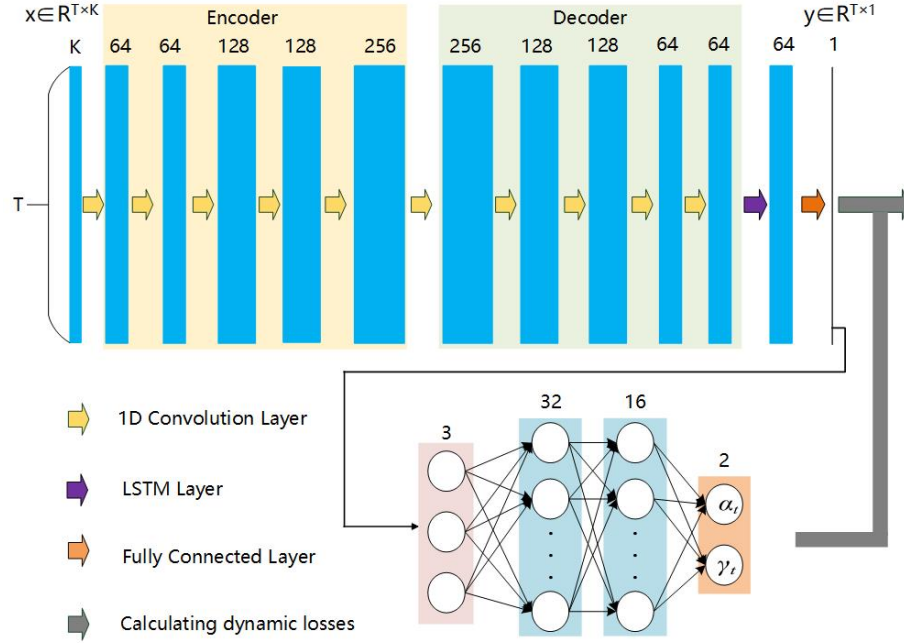


Figure 3. Architecture of the proposed CNN-LSTM model. (1) Yellow arrows: 1D convolution with ReLU activation; (2) LSTM layer; (3) fully connected layer with point-wise probability output; (4) gray arrows: dynamic loss function.

Table 2
Proportion of Missing Values for Each Characteristic of the Data

F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}
54.4%	37.1%	16.6%	5.0%	1.1%	0.3%	0.3%	0.1%	0.1%	0.3%	1.9%	4.6%	6.9%	21%	100%

Note. The features not mentioned in the table are all without missing values.

field components σ_{B_x} , σ_{B_y} , σ_{B_z} ; the proton density from moment analysis N_p ; proton density and α -particle density from nonlinear analysis $N_{p,nl}$, $N_{\alpha,nl}$; and 15 channels of proton flux between 0.3 keV and 10 keV $F_0 \sim F_{14}$. Additionally, three extra features have been calculated, including plasma- β , which quantifies the ratio of thermal pressure to magnetic pressure, the dynamic pressure $P_{\text{dyn}} = N_p V^2$, and magnetic fluctuations $\sigma_B = \sqrt{\sigma_{B_x}^2 + \sigma_{B_y}^2 + \sigma_{B_z}^2} / B$.

Data preprocessing. As shown in Table 2, the data set exhibits significant missing values in proton-flux measurements. Among the 15 channels recording proton flux, five channels demonstrate missing rates exceeding 10%, with one channel entirely absent in the observations. For time-series deep-learning applications, zero imputation provides an efficient solution to missing values by preserving temporal continuity while maintaining consistent input dimensions across samples. This approach eliminates computational complexity associated with alternative interpolation methods without compromising model stability.

Subsequently, we apply Z-score normalization (A. Demircioğlu 2024) to standardize feature scales. This process eliminates magnitude disparities between features by centering the distribution (mean=0) and scaling to unit variance (standard deviation=1): $z_i = \frac{x_i - \mu}{\sigma}$, $i = 1, 2, \dots, n$, where x_i and z_i denote the raw and normalized measurements respectively, μ and σ represent the sample mean and standard deviation, and n corresponds to the total number of input samples.

2.2. Model

We denote one input sample that contains T data points with 33 features as $x \in \mathcal{R}^{T \times 33}$. T is set to 64 after experiments as it best balances performance and efficiency. Our goal is to find a mapping $f: \mathcal{R}^{T \times 33} \rightarrow \mathcal{R}^{T \times 1}$. That is, our model assigns a label to each data point from a segmentation perspective, where “0” denotes not belonging to an ICME event, and “1” indicates belonging to an ICME event. A series of consecutive data points labeled as “1” are identified as an ICME event.

The structure of the proposed model is shown in Figure 3, which can be divided into a CNN component, an LSTM component and a dynamic loss function component:

1. **CNN component.** The CNN component consists of two parts: the encoder and the decoder. The encoder highlighted in yellow in Figure 3 is composed of five convolutional layers with channel sizes of 64, 64, 128, 128, and 256, respectively. By employing the same number of filters in two consecutive convolutional layers, the model’s feature representation capability is strengthened. Although the pooling operation can reduce computational costs and memory usage, it will lead to the loss of crucial detailed information, so the pooling operation has not been adopted. The decoder, indicated in the light green area in Figure 3, is similar to the encoder and also comprises five convolutional layers with the number of channels gradually halving each step from 256 to 64. By symmetrically designing the encoder

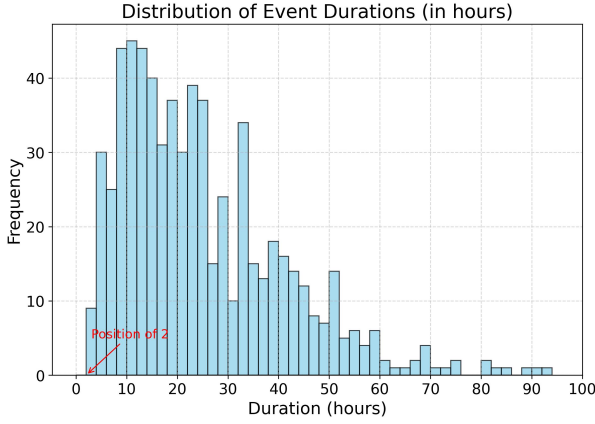


Figure 4. Histogram of time intervals between adjacent ICME events with 2 hr resolution.

Table 3
Dichotomous Confusion Matrix

	Predicted ICME	Predicted Non-ICME
ICME	TP	FN
Non-ICME	FP	TN

and decoder, the network can maintain consistency in the feature extraction and reconstruction process, allowing for more efficient capture of critical features in the input data X and achieving accurate reconstruction.

2. *LSTM component.* An LSTM layer with 64 hidden units was integrated between the decoder stage and the fully connected layer. LSTM employs a set of gating mechanisms—input gate, forget gate, and output gate—to dynamically manage information flow (S. Hochreiter & J. Schmidhuber 1997). The input/output gate selectively integrate new inputs while pruning irrelevant historical information, while the output gate control feature controls propagation to subsequent layers. Through iterative state updates, the hidden state at each time step synthesizes multiscale temporal patterns, enabling robust capture of long-range dependencies critical for ICME detection. The subsequent fully connected layer outputs a $T \times 1$ probability vector, where each element quantifies the likelihood of an ICME event occurrence at the corresponding data point.
3. *Dynamic loss function component.* Given that data points belonging to ICME events account for a very small proportion of the total data, sample class imbalance becomes a significant issue. To tackle this problem effectively, we implement a dynamically optimized focal loss (T.-Y. Lin et al. 2017):

$$L_f(p_t) = -\alpha_t(1 - p_t)^{\gamma_t} \log(p_t) - (1 - \alpha_t)p_t^{\gamma_t} \log(1 - p_t),$$

where α_t governs the significance of each class in loss computation, and γ_t fine-tunes the focus of focal loss. Inspired by L. Wu et al. (2018)’s teacher–student paradigm, we design a three-layer neural network as the teacher module. The input to the network consists of the F1 score, precision, and recall from the validation set, and the output corresponds to the two parameters α_t and

Table 4
Selection for Ratio of Positive to Negative Samples during Training

Ratio	R	P	F1 Score
0.5	0.8522	0.7232	0.7824
0.6	0.8304	0.7403	0.7828
0.7	0.8217	0.8043	0.8129
0.8	0.8522	0.7128	0.7789
0.9	0.4609	0.9550	0.6217
1.0	0.9957	0.1730	0.2947

γ_t required for the loss function. Through iterative feedback, the teacher network dynamically adjusts α_t and γ_t (final values 0.24 and 0.7) based on model performance, analogous to pedagogical adaptation where educators refine teaching strategies according to student progress.

2.3. Correction Strategy

As shown in Figure 4, our statistical analysis of the data reveals that ICME events typically last for at least 2 hr, with each bar in the histogram representing a duration of 2 hr. Therefore, we employ the following strategies, as shown at the bottom of Figure 2, to enhance the initial results generated by the model:

1. Assign a label of 0 to consecutive data points with durations less than 2 hr.
2. Consider adjacent ICME events with time differences less than 2 hr as the same ICME event.

3. Experiments and Results

In this section, we introduce the experimental setup and the experimental results to demonstrate the validity and reliability of the proposed model. The data and code is available in GitHub at <https://github.com/ht426/Automatic-detection-of-ICMEs>. The archival version used with this work is available on Zenodo at doi:10.5281/zenodo.15489068 (J. Li & H. Deng 2025).

3.1. Setup of the Experiment

In this study, an event is identified as an ICME event when the labels of at least 10 consecutive data points are marked as 1. Intersection over ground truth (IoG) was chosen to ascertain whether a predicted ICME event is a true positive (TP), with the formula as follows:

$$\text{IoG}_{ij} = \frac{\text{duration}(X_i \cap Y_j)}{\text{duration}(Y_j)}, \quad (1)$$

where X_i and Y_j represent the i th predicted ICME event and the j th real ICME event, respectively, and $\text{duration}(\cdot)$ denotes the duration of the event.

ICME detection is prone to two types of errors: false positives (FPs) and false negatives (FNs). FP occur when a non-ICME event is mistakenly predicted as an ICME event, whereas FN happen when an ICME event is inaccurately predicted as a non-ICME event, as depicted in Table 3. Precision (P) and recall (R) are chosen to measure the

Table 5
Comparison of Detection Results with Different Models

Model	Parameters	Time	R	P	F1 Score
CNN (G. Nguyen et al. 2019)	0.01M	>400 hr	0.7783	0.7512	0.7650
U-net (O. Ronneberger et al. 2015)	9.77M	4.5 hr	0.8217	0.6974	0.7535
AttU-net (O. Oktay et al. 2018)	11.51M	5.5 hr	0.8000	0.7244	0.7603
U-net++ (Z. Zhou et al. 2018)	7.11M	4.2 hr	0.8435	0.7293	0.7823
U-net+++ (H. Huang et al. 2020)	25.14M	11.4 hr	0.7391	0.8019	0.7692
RU-net (J. Chen et al. 2022)	13.61M	5.6 hr	0.7739	0.8318	0.8018
CNN-LSTM (ours)	0.61M	4.6 hr	0.8217	0.8043	0.8129

Note. The bold formatting indicates that the model achieves the best performance on the current metric. M in column (2) indicates million ($\times 10^6$).

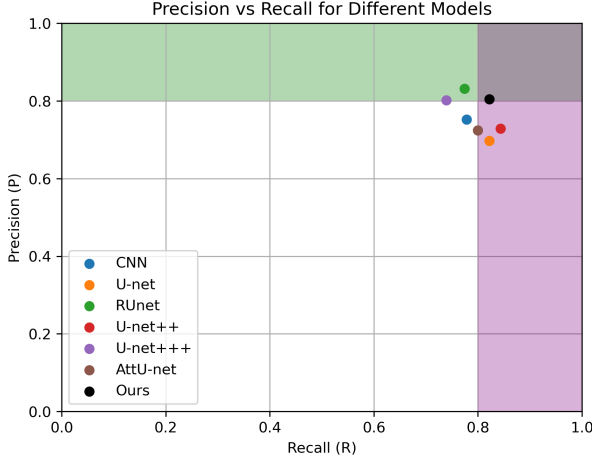


Figure 5. Precision–recall comparison of different models. Green = high-precision regime ($P > 0.80$), purple = high-recall regime ($R > 0.80$); colored dots denote candidate models.

performance of the model, with the expressions as follows:

$$R = \frac{TP}{TP + FN}, \quad (2)$$

$$P = \frac{TP}{TP + FP}. \quad (3)$$

Generally, a higher recall implies a lower precision, and vice versa. To provide a comprehensive assessment, we also take into account the F1 score, which depicts the combined performance:

$$F1 = \frac{2 \times R \times P}{R + P}. \quad (4)$$

The data sets are partitioned chronologically to cover the entire solar activity cycle: data from 1998 to 2009, 1997, and 2010 to 2015 are allocated as the training, validation, and test sets, respectively. This division yields 410 ICME events in the training set, 13 events for validation, and 230 events in the test set.

According to statistics, the ratio of positive to negative samples in the data set is approximately 1:10, affecting the attention given to positive samples. To mitigate the degradation in model performance caused by the extreme imbalance between positive and negative samples, we set the ratio of positive to negative samples in each training batch. As shown in Table 4, experiments have demonstrated that the model

performs optimally when the ratio is set to 0.7, which aligns with the conclusion drawn by J. Chen et al. (2022).

The Adam optimizer (D. P. Kingma & J. Ba 2014) with a learning rate of 0.0001 is used to minimize the focal loss. The experiments are done on a computer with a 16 vCPU Intel(R) Xeon(R) Gold 6430 CPU and an NVIDIA RTX 4090 (24 GB) GPU.

3.2. Results

To systematically evaluate the effectiveness of the proposed pipeline, we conducted a comprehensive comparative analysis with six established deep-learning models: CNN (G. Nguyen et al. 2019), U-net (O. Ronneberger et al. 2015), AttU-net (O. Oktay et al. 2018), U-net++ (Z. Zhou et al. 2018), U-net+++ (H. Huang et al. 2020), and RU-net (J. Chen et al. 2022). All models were rigorously trained and evaluated under identical configurations to ensure comparability.

The quantitative comparison presented in Table 5 reveals three key findings. First, the CNN-LSTM achieves superior overall performance with the highest F1 score (0.8129) while maintaining computational efficiency, as evidenced by its relatively small parameter count (0.61×10^6) and moderate training time (4.6 hr). Second, model specialization trends emerge: U-net+++ attains peak recall (0.8435), whereas RU-net demonstrates optimal precision (0.8318). Third, the CNN-LSTM uniquely balances these competing metrics, achieving both precision and recall above the 0.8 threshold that we define as high performance for both metrics. This balanced performance is visually confirmed in Figure 5, where the CNN-LSTM occupies the critical intersection region of high precision (>0.8) and high recall (>0.8). Other models exhibit distinct trade-offs: U-net+++ sacrifices precision (0.7293) for enhanced recall, while RU-net prioritizes precision at a slight recall cost (0.7739). These comparative results systematically demonstrate our model's advantage in maintaining metric equilibrium without specialized tuning.

Finally, Figure 6 provides validation of our correction strategy's effectiveness, contrasting raw detection outputs with postprocessed results.

4. Discussion

In this section, we conduct a comprehensive comparison between our proposed approach and the baseline model. Subsequently, we perform a detailed statistical analysis of the obtained results.

The proposed CNN-LSTM detection model synergizes the CNN architecture from G. Nguyen et al. (2019) and the time-series segmentation strategy from J. Chen et al. (2022).

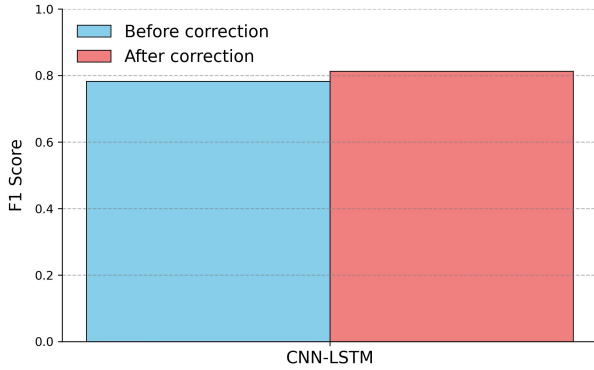


Figure 6. Comparison of F1 score before and after the application of the correction strategy.

Table 6
Performance Comparison under 10 minutes Averaging

Model	<i>R</i>	<i>P</i>	F1 Score
CNN (G. Nguyen et al. 2019)	0.7304	0.7602	0.7450
RU-net (J. Chen et al. 2022)	0.7609	0.7202	0.7400
CNN-LSTM (ours)	0.8408	0.6655	0.7429

Note. The bold formatting indicates that the model achieves the best performance on the current metric.

Compared with G. Nguyen et al. (2019), our model adopts an end-to-end framework that integrates LSTM layers to address CNN limitations in modeling long-term temporal dependencies. This enhancement yields a 6.79% improvement in F1 score while reducing training time by 94.5% (from 83.5 hr to 4.6 hr). When compared with Ru-net (J. Chen et al. 2022), our model achieves higher parameter efficiency while delivering a 1.11% higher F1 score than Ru-net, demonstrating superior computational efficiency without compromising accuracy.

To evaluate how different missing data-preprocessing methods affect detection performance, we calculate the average of available data points within each 10 minutes time window as proposed by G. Nguyen et al. (2019), rather than using zero-padding. As shown in Table 6, it can be observed that the F1 scores of the three methods are similar, yet our method has the highest recall rate. However, we are aware that in the prediction task, the consequences of missed detections are more severe than false detections. Additionally, it can be seen that the F1 scores of all three methods have decreased. Among them, the decrease of G. Nguyen et al. (2019) is the smallest, while the decrease of our method is the largest. This also indicates that the data-preprocessing method of filling with zeros is more compatible with our method.

The test set contains a total of 230 ICME events. Our model detected 235 events, including 189 TP events. Figure 7 displays the cumulative probabilities of correctly predicted data points within each ICME event as a function of time shift relative to the nearest predicted ICME events. The size of the dots corresponds to the duration of ICME events, with larger dots indicating longer durations. The solid line marks the minimum probability threshold for TP events. Notably, the range of the total probability spans from 13 to 1543, indicating a relatively wide variation, likely reflecting the positive correlation between the durations of TPs and the pattern of ICMEs (see Figures 8(a) and (c)). Using the solid line as a

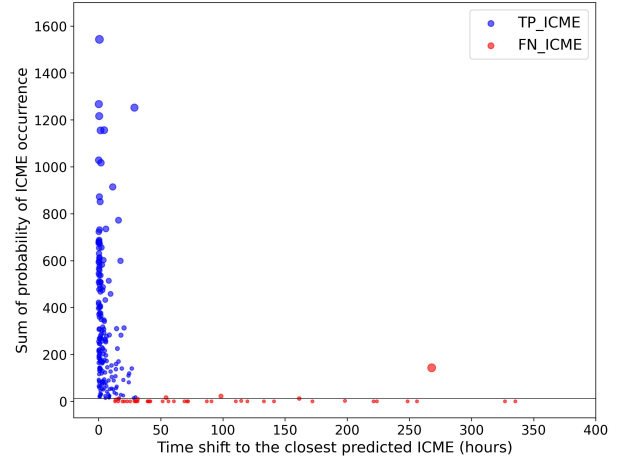


Figure 7. Cumulative probabilities of correctly predicted data points within each ICME event vs. temporal shift to the closest predicted ICME event. Blue dots and red dots stand for TPs and FNs, respectively, with dot size corresponding to the duration of the ICME event; the black line indicates the minimum temporal shift of the TPs.

divider, the FNs events fall into two categories. Most FNs are located below the solid line, where they are weak in both duration and pattern, making them difficult to detect. Similarly, FNs located above the solid line are either too short in duration or too weak in pattern to be detected (see Figure 8(e)). Alongside magnetic field intensity patterns, we annotated corresponding predicted labels (see Figures 8(b), (d), and (f)). Crucially, FNs detected by our model appear distant from predicted ICME events and correspond to highly isolated cases—a finding consistent with studies by G. Nguyen et al. (2019) and J. Chen et al. (2022).

We conducted a systematic analysis of the durations and probability sum distributions for both TPs and FPs, as shown in Figure 9. The comparative analysis reveals two key patterns. First, the durations of the FPs are predominantly less than 10 hr, with their probability sums being significantly lower than those of most TPs. This duration–probability correlation suggests that brief, low-confidence detections are more prone to misclassification. Second, we identified an anomalous subgroup of FPs exhibiting extended durations and elevated probability sums comparable to confirmed ICMEs. These “ICME-like” false alarms present particularly challenging cases for discrimination, as their temporal and probabilistic characteristics overlap substantially with authentic events. This implies that developing multidimensional discrimination criteria—combining duration thresholds with probability sum ranges—could significantly improve detection accuracy.

Furthermore, to validate the physical reliability of our model, we analyze the agreement in physical characteristics between predicted and observed ICME events. As shown in Figure 10, the statistical distributions of three key parameters—event duration, magnetic field, and proton thermal velocity—are comparatively shown, with the left column representing simulation results and the right column displaying observational data. Notably, while minor discrepancies exist in ICME durations, the overall profiles for magnetic field and thermal velocity demonstrate remarkable congruence, thereby substantiating the physical plausibility of our model’s outputs when compared with established ICME catalogs.

In order to evaluate the robustness of the model, we conduct comprehensive evaluations across multiple data set

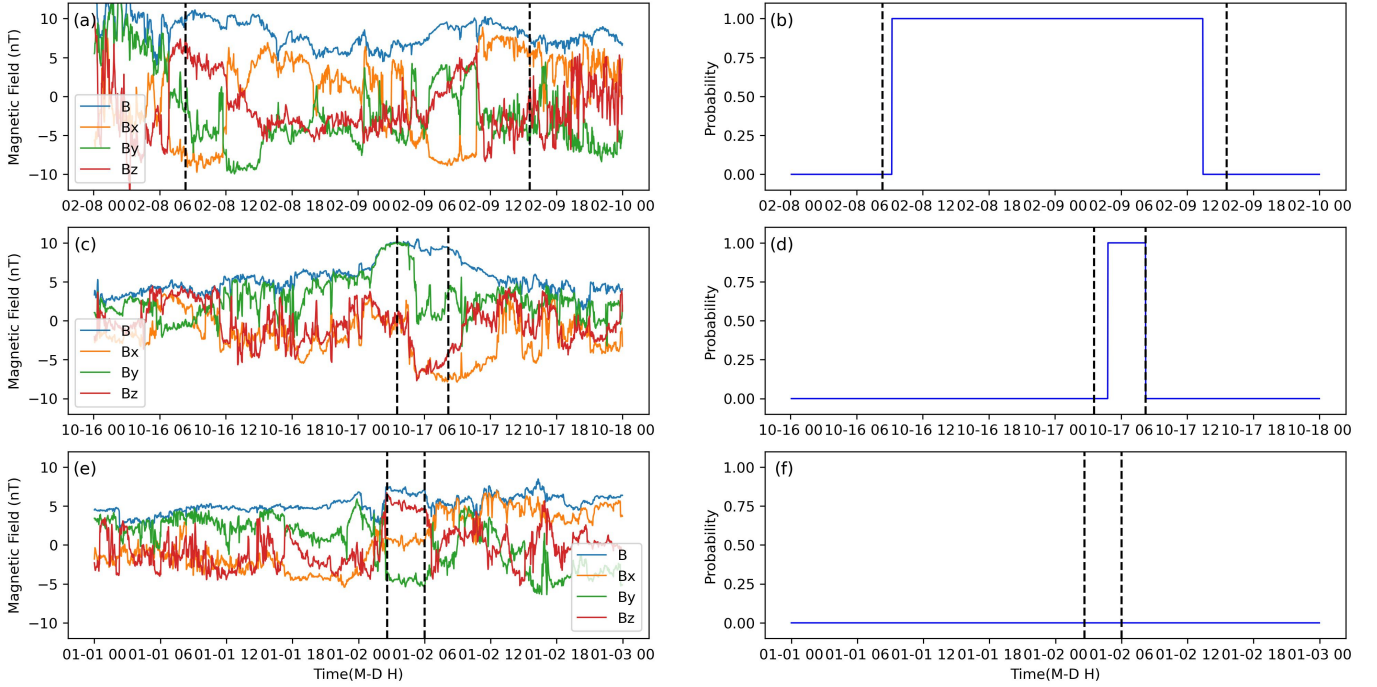


Figure 8. Magnetic field profiles and detection results of three ICME events: (a), (b) successful long-duration case; (c), (d) successful short-duration case; (e), (f) failed short-duration case, with vertical dotted lines marking the boundaries of ICME events.

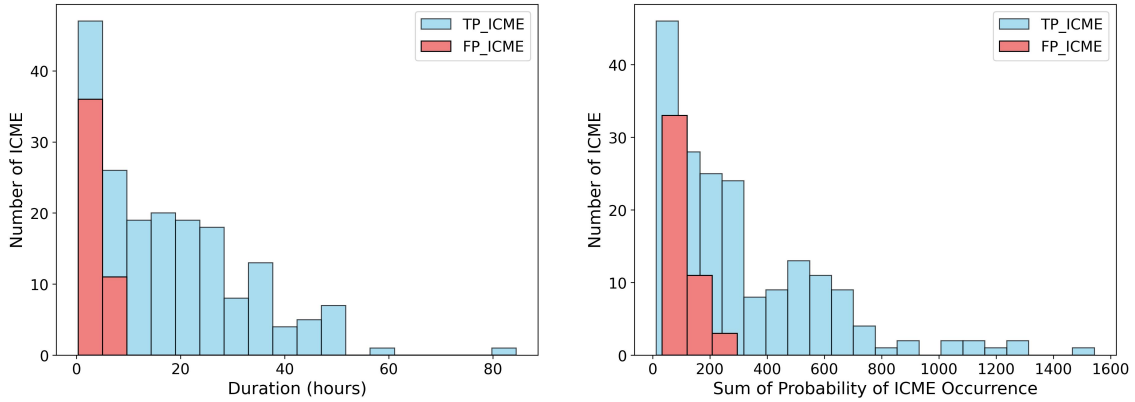


Figure 9. Duration and probability sum histograms for both TP and FP.

configurations with distinct chronological divisions for training, validation, and testing. Three temporal splits are implemented:

1. Training: 1998 January–2009 December. Validation: 1997 October–1997 December. Testing: 2010 January–2015 December.
2. Training: 2004 April–2015 December. Validation: 2004 January–2004 March. Testing: 1997 October–2003 December.
3. Training: 1997 October–2003 December and 2010 April–2015 December. Validation: 2010 January–2010 March. Testing: 2004 January–2009 December.

The method of data set division is consistent with that in G. Nguyen et al. (2019) and J. Chen et al. (2022), and the results are shown in Table 7. It can be seen that our model is greatly affected by the training data. On the other hand, we evaluate the performance of the model when partial features

are selected for training, and the results are shown in Table 8. The model learned with partial features performs slightly worse than the model learned with all features. Meanwhile, it can be established that the two features of magnetic field and proton flux are of great importance for the automatic detection of ICME events.

5. Conclusion

A CNN-LSTM model with dynamic loss for ICME detection is proposed in this paper. The main contributions are summarized as follows:

1. *Efficiency.* It outperforms the Ru-net (J. Chen et al. 2022) by reducing detection time by 1 hr and boosting the F1 score by 1.11%, demonstrating superior speed and accuracy in detection tasks.
2. *Robustness and generalization.* By incorporating LSTM, it effectively captures long-term temporal dependencies

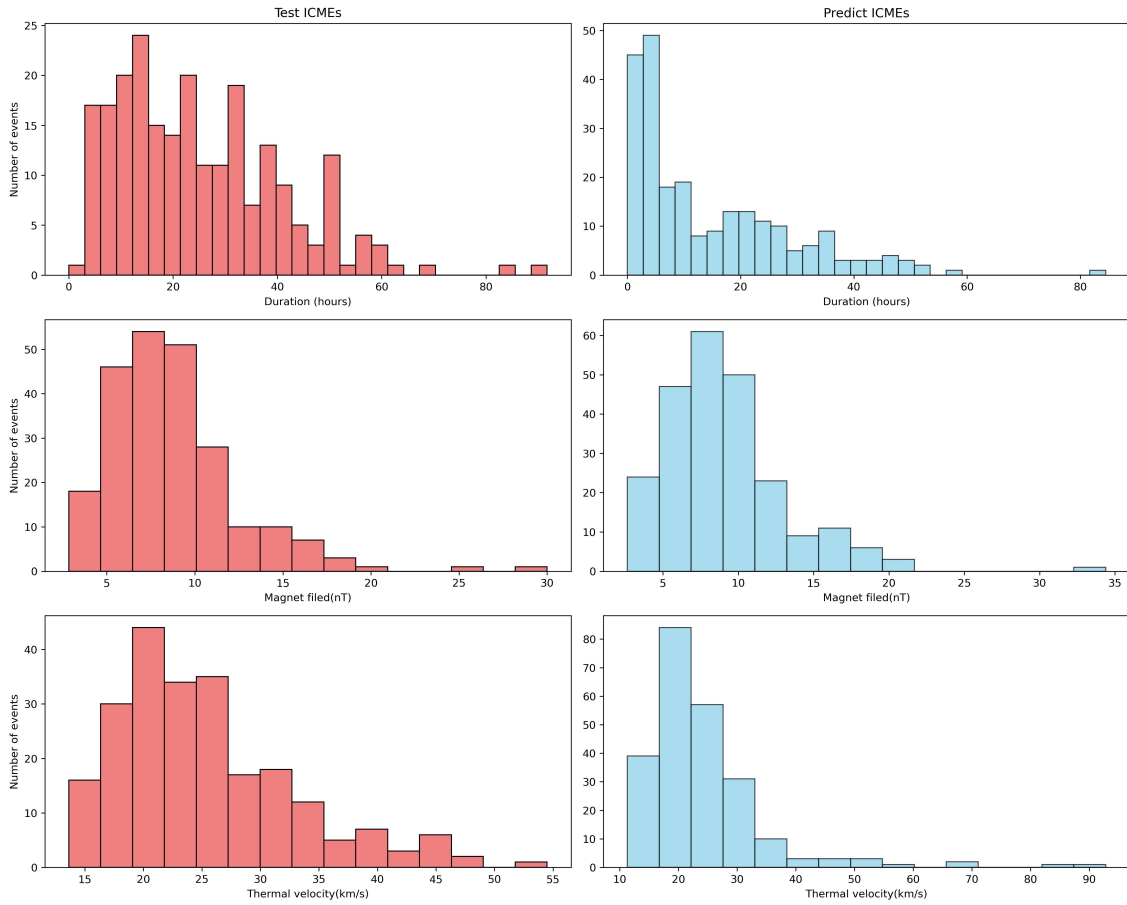


Figure 10. Histograms of the duration, magnetic field, and thermal velocity distributions for both test and predicted ICMEs.

Table 7

Performance of Our Model under Different Data Partitions

Training Period	R	P	F1 Score
1998.01–2010.01	0.8217	0.8043	0.8129
2004.04–2015.12	0.8500	0.6071	0.7083
1997.10–2003.12 and 2010.04–2015.12	0.8020	0.4208	0.5519

Table 8

Performance of Our Model Using Different Combinations of Features

Data Set Features	R	P	F1 Score
All	0.8217	0.8043	0.8129
B_x, B_y, B_z, β and proton fluxes	0.7174	0.8730	0.7876
Proton fluxes	0.5000	0.8984	0.6425
B, B_x, B_y, B_z, β	0.4783	0.9402	0.6340
B, B_x, B_y, B_z	0.3695	0.9671	0.5397

within time-series data, surpassing the time-handling capacities of prior models.

Building on the ideas presented in this paper, several key challenges remain. Extracting more insights from the data set is crucial for developing better ICME prediction models. However, due to limitations in data collection, ICME data often has missing or incomplete features, and discarding them would lead to significant losses. Additionally, ICME events are rare, creating a data imbalance, which poses challenges for

effective modeling. Future research should focus on data analysis techniques tailored to the unique characteristics of solar flare data sets.

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